EXPERIMENT 1

Determination of the molar mass by freezing point depression

Goal

The goal of this experiment is to measure the freezing point depression of a solvent and to use this depression to measure the molecular weight of a solute.

Materials

- ☐ A ring stand and two iron rings with a wire gauze
- ☐ two or three jumbo text tubes with stoppers
- ☐ A series of 50mL, 100mL, 250mL and 600mL beaker
- ☐ Cyclohexane
- ☐ 20mL pipet
- ☐ Ice
- ☐ Perforated stoppers with stirring wire and thermometer

Background

During winter, salt is poured over the street to melt the ice on the floor. General knowledge says that the salt melts the ice, while some venture to claim that the salt lowers the freezing temperature of the water. Both assertions might be in the right direction but are, if not false, at least incorrect. When salt is mixed with water, there is no water or salt anymore. Instead, there is a *solution* of salt in water. Since the freezing point of the solution is lower than that of the pure water, the solution is no longer solid at outdoor temperatures. Notice that there is no need to state the nature of the solution or the solute because it is irrelevant. The depression of the freezing point is not a property specific to the salt, (i.e. the sodium chloride, NaCl), but rather a general effect of any solute. Salt is used because it is widely available, dissolves fast, and is inexpensive.

Freezing point depression

The formula governing the change (Δ) in the freezing point (T_f) is given by:

$$\Delta T_f = - K_f m$$

where m is the concentration in terms of *molality* and K_f is the *molal freezing-point depression constant.* K_f is characteristic of the solvent being experimentally determined. Remember that molality m refers to the moles of solute concerning the kilograms of solvent. ΔT_f is the difference between the freezing point of the pure solvent ($T_f^{solvent}$) and that of the solution ($T_f^{solution}$). The solvent used in this experiment will be *cyclohexane*, a very *volatile* organic solvent. Special precautions must be taken to avoid evaporation of the solvent. K_f for cylohexane is 20.1 °C/m. The solute will be an unknown compound. Hence you will not know its molar mass and therefore the molarity will not be explicitly known here.

Example

The freezing point of pure benzene is 5.50°C. A solution is prepared by dissolving 0.450 g of an unknown substance in 27.3 *g of benzene. The new freezing point is determined to be 4.18 °C. What is the molar mass of the unknown substance? The freezing point constant, K_f, for benzene is 5.12 °C/m.*

Answer: calculate the concentration first:

$$\Delta T_f = K_f \times m \qquad m = \frac{\Delta T_f}{K_f} \qquad m = \frac{5.50°C - 4.18°C}{5.12°C\frac{kg}{mol}} = 0.258\frac{mol}{kg}$$

then, calculate the molar mass:

$$0.258\frac{mol}{kg} \times 27.3g \times \frac{1kg}{1000g} = 7.04 \times 10^{-3} mol \qquad \text{molar mass} = \frac{0.450g}{7.04 \times 10^{-3} mol} = 63.9\frac{g}{mol}$$

The experiment

The experiment today is divided into the following parts: finding the freezing point of the pure solvent, finding the freezing point of the solution, finding the freezing point after increasing the concentration, and solving the molar mass of the solute.

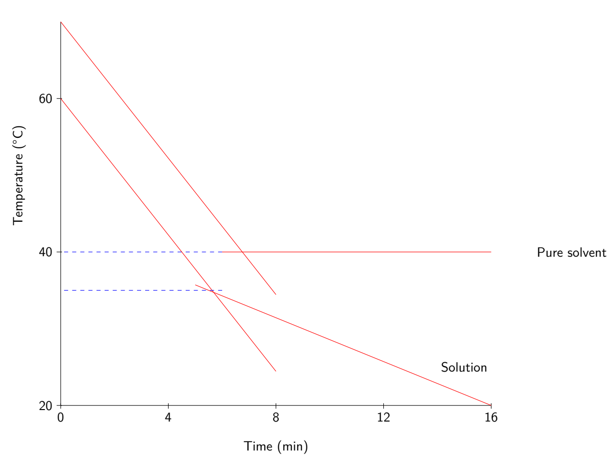
Cooling plots

During a phase change, all the energy transferred is used to reorganize the molecules and none is used to change the temperature of the substance. A liquid at a higher temperature than its surroundings gives up the heat and its temperature falls until it reaches the freezing point. The liquid continues to give up heat to the surroundings, but its temperature remains constant. Only when all the liquid is frozen will the temperature start to fall again.

If one plots temperature vs. time for the freezing of the pure liquid, a negative slope line will represent the cooling down, while the flat horizontal line (plateau) corresponds to the freezing process. These types of plots are called cooling plots. Notice that the transition from the cooling line to the horizontal line is not sharp. For that reason, the best way to determine the freezing point is to draw a straight line fitting most of the cooling points and another straight line fitting most of the cooling points, while finding the crossing point between two lines. You can measure the freezing point as the temperature measurement of this crossing point. For a pure solvent, the freezing point should be near the temperature corresponding to the plateau. This is the graphical method to find freezing points. In the case of a solution, this constant-temperature plateau does not exist anymore. In contrast, when a solution freezes its temperature decreases with time. However, the same graphical method can be used to obtain the freezing point of a solution. Mixing the solution well is critical to obtaining a good cooling plot. If you do not mix the solution well you will obtain a series of steps when the solution freezes that will not allow you to measure an accurate freezing point.

Finding the freezing point of the pure solvent.

In this part of the experiment, you will obtain the cooling plot of the pure solvent, cyclohexane. Using the graphical method you will obtain the freezing point as an average between two replications. You must plot the results carefully to obtain accurate results.



Finding the freezing point of a solution.

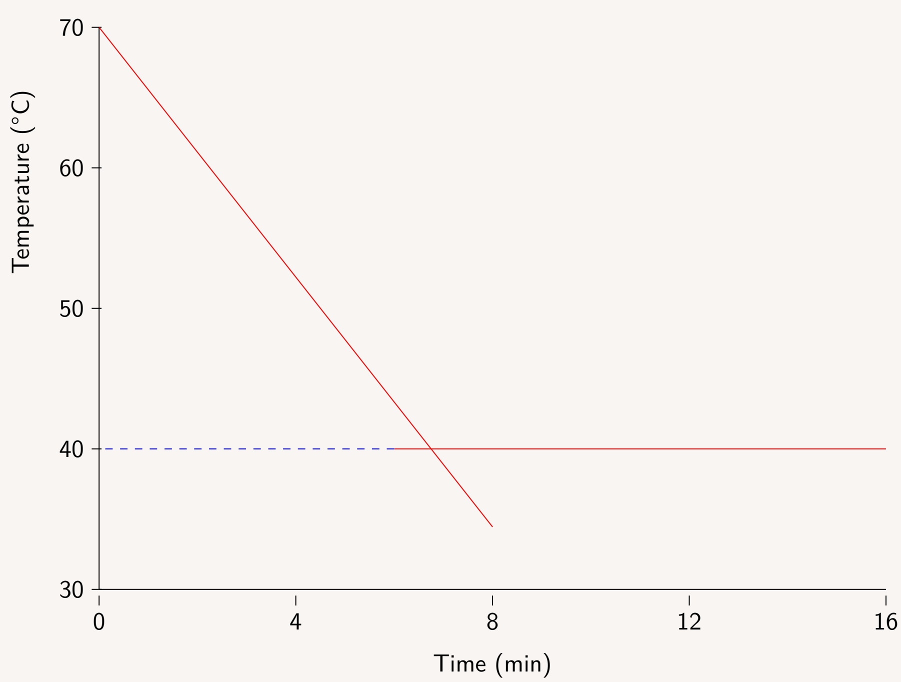
In this part of the experiment, you will obtain the cooling plot of a solution made of cyclohexane and an unknown solute. Using the graphical method you will obtain the freezing point. You will work with two different concentrations and will carry out one replication per concentration. This part aims to calculate the molar mass of the solute.

Example

Use the date replorted below to graph a cooling plot, and use the graphical method to calculate the freezing point of the substance. Is this a pure substance or a solution?

t (min)	T (°C)	t (min)	T (°C)
1	65.6	9	40
2	61.6	10	40
3	56.7	11	40
4	52.2	12	40
5	47.8	13	40
6	43.3	14	40
7	38.9	15	40
8	40	16	40

Answer: Based on the plot below, we have that at early times the liquid is being cooled down and its temperature decreases with time. At late times, the liquiz freezes and its temperature reaches a plateau. The graphical method to obtain freezing points consist on ploting a line for early times and another line for late times. These two lines intersect in a point, and the temperature for this point is the freezing point. As the temperature reaches a platea, we are dealing here with a pure substance.



Procedure

Preparing the setup.

☐ *Step 1:* – Prepare the set-up to cool down the solution. Set up a ring stand and an iron ring with wire gauze on top. The height of the platform will normally be very low.

☐ *Step 2:* – Use a second, larger ring above the platform as protection. Attach a small clamp above the second iron ring.

☐ *Step 3:* – Put ice with some water in a 500 mL beaker and set it on the platform. This will be your cool bath. Water helps create a homogeneous bath.

Good Lab Practice

✍ Never add the liquid or solid to be weighed to the container while on the scale. Take the container *off* the scale prior to adding the product. A spill on the scale will ruin both your measurement and the scale.

Preparing a cyclohexane sample.

☐ *Step 1:* – Find 100mL and 50mL beakers and a 20 mL pipet.

☐ *Step 2:* – Obtain a large test tube that is clean and dry, a suitable rubber stopper, and a 100 mL beaker. The beaker will be used to hold the test tube vertically. Get the mass of the test tube with stopper and the beaker in the scale and write it down in the Results section.

☐ *Step 3:* – Pour no more than 25 mL of cyclohexane into the 50 mL beaker. Using the pipet, transfer exactly 20.0 mL of cyclohexane from the 50 mL beaker to the test tube and close it with the stopper immediately. Mind that as the liquid is volatile the measured volume will not be accurate. Using the scale get the mass of the test tube with the cyclohexane, the stopper, and the beaker. Make sure you do not weigh the thermometer or the stirring system.

☐ *Step 4:* – Get the mass of the filled-test tube with stopper and the beaker in the scale and write it down in the Results section.

Part A. Freezing point of the pure solvent.

☐ *Step 1:* – Insert the test tube into the ice and hold it vertically using the clamp. Make sure all liquid is fully submerged in the cool bath. Find the perforated stoppers with the stirring wire and the thermometer. The round wire end must encircle the thermometer probe. Exchange the stopper on the test tube with this gadget.

☐ *Step 2:* – As soon as the temperature is 15°C (not lower) start stirring the solvent with the wire and record the temperature every 15 seconds. Use all the digits given by the thermometer down to the tenth of the degree Celsius.

☐ *Step 3:* – Start plotting the results on the graph. Plotting temperature (in °C) in the vertical axis and time in the horizontal axis. It is critical that you graph as you measure.

☐ *Step 4:* – Stop the experiment when the temperature starts dropping after several minutes of having a constant temperature. Do not stop until the temperature reaches at least 7°C.

☐ *Step 5:* – Do not proceed unless you show your instructor the plot.

☐ *Step 6:* – Now you will repeat the measurement one more time.

☐ *Step 7:* – Lift the test tube with the clamp above the ice and allow the cyclohexane to melt. Avoid using your hands to accelerate the process. Before the cyclohexane gets to 15°C get a second measurement by repeating the step above.

Good Lab Practice

- ✍ Stirr, and do not shake. It is very important to stir the cyclohexane constantly and gently until it is completely frozen.
- ✍ Make sure the wire stays around the thermometer and that the latter is centered in the test tube.
- ✍ The thermometer should not touch the walls of the test tube.
- ✍ A bad stirring method will lead to a partial freezing and the graph will not show straight lines.

Part B. Freezing point of the less concentrated solution.

☐ *Step 1:* – Pick up an unknown solute. You have to carry out Parts B and C at the same time.

☐ *Step 2:* – The same cyclohexane can be used in Part B. However, if you work in teams you can prepare another cyclohexane sample and carry out work in parallel.

☐ *Step 3:* – If you use the sample from part A, lift the test tube with the clamp above the ice and allow the cyclohexane to melt. Avoid using your hands to accelerate the process. During the next steps be careful to avoid the cyclohexane's temperature from getting above 15°C.

☐ *Step 4:* – If you prepare a new cyclohexane sample, see the steps above.

☐ *Step 5:* – Tare the scale with the weighing boat and obtain between 0.10 g and 0.11 g of the unknown solute. Remove the boat with the solute, tare the empty scale, and record the mass of the weighing boat together with the solute. Use the balance of maximum precision.

☐ *Step 6:* – Carefully add the solute into the test tube with the pure solvent. Make sure no solute is spilled. There is no need to add all the solute from the weighing boat, the amount added will be calculated by difference.

☐ *Step 7:* – Weight the weighing boat with the solute leftovers. Record the mass on the results page.

☐ *Step 8:* – Mix the solution well and insert the test tube in the ice while holding it vertically using the clamp.

☐ *Step 9:* – Start stirring the solution with the wire and record the temperature every 15 seconds. This time the temperature should not reach constant. Stop the experiment when the temperature reaches 1°C.

Part C. Freezing point of the more concentrated solution.

☐ *Step 1:* – Lift the test tube with the clamp above the ice and allow the cyclohexane to melt. Avoid using your hands to accelerate the process. During the next steps be careful to avoid the cyclohexane getting above 15°C.

☐ *Step 2:* – The same boat can be reused without cleaning. Tare the scale with the boat inside. Obtain between 0.24 g and 0.25 g of the unknown solute and record the mass of the weighing boat together with the solute using maximum precision.

☐ *Step 3:* – Add carefully the solute into the test tube with a fresh new 20mL sample of the pure solvent. Make sure no solute is spilled. There is no need to add all the solute from the weighing boat, the amount added will be calculated by difference.

☐ *Step 4:* – Weight the weighing boat with the solute leftovers. Record the mass on the results page.

☐ *Step 5:* – Mix the solution well and insert the test tube in the ice while holding it vertically using the clamp.

☐ *Step 6:* – Start stirring the solution with the wire and record the temperature every 15 seconds. This time the temperature should not reach a constant. Stop the experiment when the temperature reaches 0.5°C.

Calculations

(1) This is the mass of the empty test tube with a cork and placed in a beaker.

(2) This is the mass of the test tube filled with cyclohexane, with a cork, and placed in a beaker.

(3) This is the mass of solvent, m_{liquid}:

$$m_{solvent} = (2) - (1)$$

(4) This is the volume of cyclohexane added, where d is the density of cyclohexane:

$$v_{solvent} = \frac{m_{liquid}}{d} = \frac{(3)}{0.779}$$

(5) These are two replicates for the freezing point of the solvent obtained through the graphical method.

(6) This is the freezing point of the solvent.

(7) This is the mass of the weighing boat with solute before adding the solute to the solvent.

(8) This is the mass of the weighing boat with solute after adding the solute to the solvent.

(9) This is the mass of solute:

$$m_{solute} = (7) - (8)$$

(10) These are the freezing points of two solutions.

(11) These are the values of the freezing points depression:

$$\Delta T_f = (10) - (6)$$

(12) These is the molar mass of the solute:

$$\text{MW} = -\frac{K_f \cdot m_{solute}}{\Delta T_f \cdot m_{solvent}} \times 1000 = -\frac{20.1 \cdot (9)}{(11) \cdot (3)} \times 1000$$

STUDENT INFO

Name: ____________________ Date: __________

Pre-lab Questions

Determination of the molar mass by freezing point depression

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1. We measured the change of the freezing point of a solution with concentration at a fixed temperature. Use the data below to experimentally compute $i \cdot k_f$ for the solution and the freezing point of the pure solvent. Assume $i \cdot k_b$ does not change for the molarity range given.

m	T_f (°C)
0.010	5.448
0.011	5.444
0.012	5.438
0.013	5.433
0.014	5.428

2. From the following solutions: (i) 0.1m KCl in water (ii) 0.2m NaCl in water (iii) 0.3m KOH in water select the one with: (a) lowest freezing point (b) highest freezing point (c) highest boiling point (d) lowest boiling point (e) lowest vapor pressure (f) highest vapor pressure (g) highest osmotic pressure (h) lowest osmotic pressure

3. Use the data reported below to graph a cooling plot and use the graphical method to calculate the freezing point of the substance. Is this a pure substance or a solution?

t (min)	T (°C)	t (min)	T (°C)
1	55.6	9	30
2	51.1	10	28.6
3	46.7	11	27.1
4	42.2	12	25.7
5	37.8	13	24.3
6	33.3	14	22.86
7	28.9	15	21.6
8	31.4	16	20

(the question continues on the back)

4. What is the freezing point you calculated in the previous question?

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Results
EXPERIMENT 1

Determination of the molar mass by freezing point depression

Parts A. Freezing point of the pure solvent.

1	Mass of the empty test tube with cork and beaker:	________ g
2	Mass of the filled test tube with cork and beaker:	________ g
3	Mass of the cyclohexane:	________ g
4	Volume of the cyclohexane used:	________ mL

Trial 1				Trial 2			
Time (sec)	Temp. (°C)	Time (sec)	Temp. (°C)	Time (sec)	Temp. (°C)	Time (sec)	Temp. (°C)
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______

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5 Freezing point of the pure cyclohexane. Trial 1: __________ °C Trial 2: __________ °C

6 Mean: __________ °C

Parts B and C. Freezing point of solutions.

1	Mass of the empty test tube with cork and beaker:	________	g
2	Mass of the filled test tube with cork and beaker:	________	g
3	Mass of the cyclohexane:	________	g

Low concentration				High concentration		
	Unkown #:	________				
7	Mass of boat:	________	g	Mass of boat:	________	g
8	Mass of boat (after adding):	________	g	Mass of boat (after adding):	________	g
9	Mass of solute:	________	g	Mass of solute:	________	g

Time (sec)	Temp. (°C)	Time (sec)	Temp. (°C)	Time (sec)	Temp. (°C)	Time (sec)	Temp. (°C)
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______
______	______	______	______	______	______	______	______

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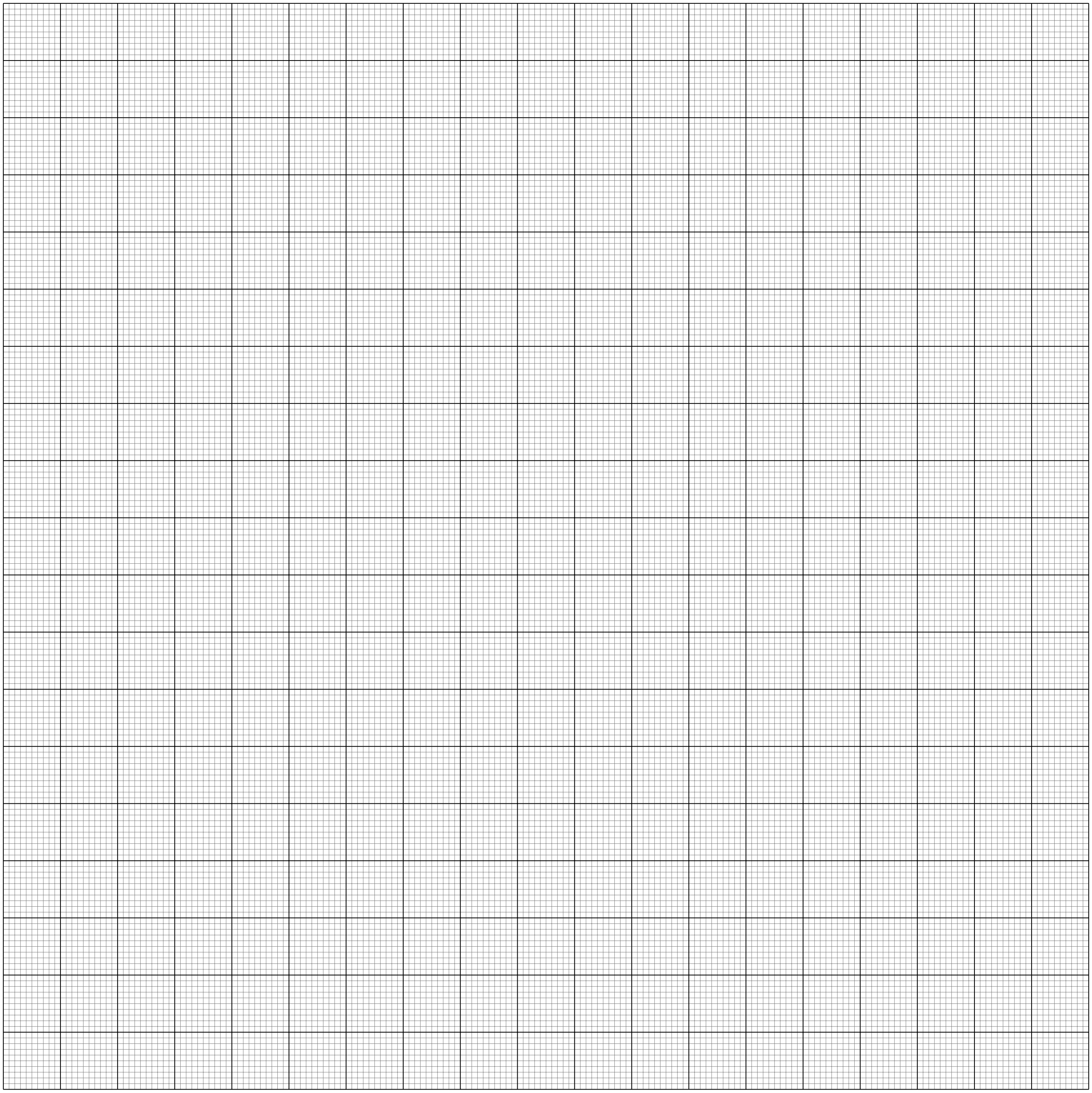
	Low concentration		High concentration	
10	Freezing point:	________ °C	Freezing point:	________ °C
11	ΔT_f:	________ °C	ΔT_f:	________ °C
12	Molar mass:	________ g/mol	Molar mass:	________ g/mol

Mean molar mass: ________ g/mol



Graph 1 Temperature (Y axis) vs. time (X axis). Replicate 1

Graph 2 Temperature (Y axis) vs. time (X axis). Replicate 2

Graph 3 Temperature (Y axis) vs. time (X axis). Low concentration.

Graph 4 Temperature (Y axis) vs. time (X axis). High concentration.

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Post-lab Questions

Determination of the molar mass by freezing point depression

1. If the empirical formula of the solute used in this experiment is C_3H_2Cl, what is the molecular formula?

2. From the following solutions: (i) 0.1m KCl in water (ii) 0.1m CaF_2 in water (iii) 0.1m $Ca_3(PO_4)_4$ in water select the one with: (a) lowest freezing point (b) highest freezing point (c) highest boiling point (d) lowest boiling point (e) lowest vapor pressure (f) highest vapor pressure (g) highest osmotic pressure (h) lowest osmotic pressure

3. From the following solutions: (i) 0.2m CH_3OH in water (ii) 0.2m NaF in water (iii) 0.2m $Ca(NO_3)_2$ in water select the one with: (a) lowest freezing point (b) highest freezing point (c) highest boiling point (d) lowest boiling point (e) lowest vapor pressure (f) highest vapor pressure (g) highest osmotic pressure (h) lowest osmotic pressure

4. Calculate the freezing point depression of a 2m I_2 solution on benzene. $T_f^{solvent}$=5.5°C and k_f=4.9°C/m.